

where the first inequalities are ensured by Theorem 2. However, the irreducibility condition $(H\kappa_{irr})$ is not satisfied by κ^{red} , so uniqueness is not ensured and Proposition 1 does not apply to guarantee that N_1^{red} and N_2^{red} are non-zero. Adapting the proof of Proposition 1 to each of the strongly connected components of the graph of κ , simply reduced to $\{v_1\}$ and $\{v_2\}$, shows that for each $i \in \{1, 2\}$, N_i^{red} is either zero everywhere or positive on $(\frac{b}{2}, +\infty)$. The normalization condition

$$\sum_{i \in \{1, 2\}} \int_0^\infty N_i^{red}(x) dx = 1$$

implies that at least for one $i \in \{1, 2\}$, N_i^{red} is non-zero. Both N_1^{red} and N_2^{red} cannot be non-zero, otherwise Lemma 2 would work (in equality (14), we need the existence of $i \neq j$ such that $N_i^{red} \neq 0$ and $N_j^{red} \neq 0$) and we could use it to prove the long-time convergence of system (20) towards a stationary state (fourth step of Theorem 4). That would contradict [6]. The only possibility is that $N_1^{red} \equiv 0$ and $N_2^{red}(x) > 0$ for $x \geq \frac{b}{2}$ or the opposite and then $\lambda = \lambda_2$ or $\lambda = \lambda_1$ respectively.

In the mixing case, the equations of system $(GF_{t,v})$ are coupled through their source term:

$$\begin{cases} \frac{\partial}{\partial t} n_1(t, x) + v_1 \frac{\partial}{\partial x} (x n_1(t, x)) + \gamma_1(x) n_1(t, x) = 4\gamma_2(2x) n_2(t, 2x), \\ \frac{\partial}{\partial t} n_2(t, x) + v_2 \frac{\partial}{\partial x} (x n_2(t, x)) + \gamma_2(x) n_2(t, x) = 4\gamma_1(2x) n_1(t, 2x), \\ n_1(0, x) = n_1^{in}(x), \quad n_2(0, x) = n_2^{in}(x), \end{cases}$$

and the irreducibility condition $(H\kappa_{irr})$, missing in the non-mixing case, is now satisfied. We can thus apply successively Theorem 3, Proposition 1 and Theorem 4 to get eigenelements $(\lambda^{irr}, N^{irr}, \phi^{irr})$ such that

$$\sum_{i=1}^2 \int_{\mathbb{R}^+} |n_i(t, x) e^{-\lambda^{irr} t} - N_i^{irr}(x)| \phi_i^{irr}(x) dx \xrightarrow{t \rightarrow \infty} 0$$

with $\lambda^{irr} > 0$, and for $i \in \{1, 2\}$, N_i^{irr} positive on $(\frac{b}{2}, +\infty)$ and $\phi_i^{irr} > 0$ on $(0, +\infty)$.

These two simple cases illustrate that the existence result (Theorem 3) holds for every probability matrix κ , in particular reducible ones. The irreducibility condition on κ comes into play to characterize the functions canceling the dissipation of entropy (Lemma 2), which then proves crucial to establish uniqueness of the steady state and convergence towards it.

3.2 Numerical illustration

Similarly to the previous subsection, we focus here on the special case of linear growth rates to illustrate the convergence result of Theorem 4. We numerically approximate and compare the long-time asymptotics in the presence and absence of mixing in feature.

We choose $M = 3$ different features, namely $\mathcal{V} = \{1, 2, 3\}$, and approximate on the grid

$$\mathcal{S}_N := \mathcal{V} \times \{x_0, \dots, x_{2N}\}, \quad x_m := 2^{\frac{-m-N}{k}}, \quad m \in \{0, \dots, 2N\}, \quad k = 200, \quad N = 2501,$$

the time-evolution of the following initial data (taken identical for all features)

$$n^{in} : (v, x) \mapsto C x^a e^{-bx^2}, \quad a = 30, \quad b = 60, \quad C \text{ s.t. } \|n^{in}\|_{L^1(\mathcal{V} \times (0, x_{2N}))} = 1,$$